

MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences
UNIST

Mathematical Analysis I & II

Course Textbook

Author

Muho Choi (Department of Computer Science and Engineering)

Email: sihyeon8240@unist.ac.kr

GitHub: <https://github.com/sihyeon8240>

Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/Math>

Feel free to open issues or pull requests for corrections.

© 2026. Muho Choi. All rights reserved.

Developed for the Department of Mathematical Sciences, UNIST.

This textbook is licensed under a **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License** (CC BY-NC-SA 4.0).

You are free to: Share, copy, and redistribute the material in any medium or format; adapt, remix, transform, and build upon the material.

Under the following terms: You must give appropriate credit; you may not use the material for commercial purposes; if you remix or transform the material, you must distribute your contributions under the same license.

Contents

1	The Number Systems	1
1.1	The Natural, Integer, and Rational Number Systems	1
1.2	Ordered Sets	2
1.3	Fields and the Real Field	5
1.4	The Extended Real Number System	10
1.5	The Complex Field	10
1.6	Euclidean Spaces	14
2	Basic Topology	16
2.1	Relations	16
2.2	Functions	17
2.3	Finite Sets	20
2.4	Countable and Uncountable Sets	27
2.5	Metric Spaces	35
2.6	Compact Sets	35
2.7	Perfect Sets	35
2.8	Connected Sets	35
3	Numerical Sequences and Series	36
3.1	Convergent Sequences	36
3.2	Subsequences	36
3.3	Cauchy Sequences	36
3.4	Upper and Lower Limits	36
3.5	Some Special Sequences	36
3.6	Series	36
3.7	Series of Nonnegative Terms	36
3.8	The Number e	36
3.9	The Root and Ratio Tests	36
3.10	Power Series	36
3.11	Summation by Parts	37
3.12	Absolute Convergence	37
3.13	Addition and Multiplication of Series	37
3.14	Rearrangements	37

4	Continuity	38
4.1	Limits of Functions	38
4.2	Continuous Functions	38
4.3	Continuity and Compactness	38
4.4	Continuity and Connectedness	38
4.5	Discontinuities	38
4.6	Monotonic Functions	38
4.7	Infinite Limits and Limits at Infinity	38
5	Differentiation	39
5.1	The Derivative of a Real Function	39
5.2	Mean Value Theorems	39
5.3	The Continuity of Derivatives	39
5.4	L'Hospital's Rule	39
5.5	Derivatives of Higher Order	39
5.6	Taylor's Theorem	39
5.7	Differentiation of Vector-valued Functions	39
6	The Riemann-Stieltjes Integral	40
6.1	Definition and Existence of the Integral	40
6.2	Properties of the Integral	40
6.3	Integration and Differentiation	40
6.4	Integration of Vector-valued Functions	40
6.5	Rectifiable Curves	40
7	Sequences and Series of Functions	41
7.1	Discussion of Main Problem	41
7.2	Uniform Convergence	41
7.3	Uniform Convergence and Continuity	41
7.4	Uniform Convergence and Integration	41
7.5	Uniform Convergence and Differentiation	41
7.6	Equicontinuous Families of Functions	41
7.7	Equicontinuous Families of Functions	41
7.8	The Stone-Weierstrass Theorem	41
8	Some Special Functions	42
8.1	Power Series	42
8.2	The Exponential and Logarithmic Functions	42
8.3	The Trigonometric Functions	42
8.4	The Algebraic Completeness of the Complex Field	42
8.5	Fourier Series	42
8.6	The Gamma Function	42

9	Functions of Several Variables	43
9.1	Linear Transformations	43
9.2	Differentiation	43
9.3	The Contraction Principle	43
9.4	The Inverse Function Theorem	43
9.5	The Implicit Function Theorem	43
9.6	The Rank Theorem	43
9.7	Determinants	43
9.8	Derivatives of Higher Order	43
9.9	Differentiation of Integrals	43
10	Integration of Differential Forms	44
10.1	Integration	44
10.2	Primitive Mappings	44
10.3	Partitions of Unity	44
10.4	Change of Variables	44
10.5	Differential Forms	44
10.6	Simplexes and Chains	44
10.7	Stokes' Theorem	44
10.8	Closed Forms and Exact Forms	44
10.9	Vector Analysis	44
11	The Lebesgue Theory	45
11.1	Set Functions	45
11.2	Construction of the Lebesgue Measure	45
11.3	Measure Spaces	45
11.4	Measurable Functions	45
11.5	Simple Functions	45
11.6	Integration	45
11.7	Comparison with the Riemann Integral	45
11.8	Integration of Complex Functions	45
11.9	Functions of Class $\mathcal{L}^2$	45

Chapter 1

The Number Systems

1.1 The Natural, Integer, and Rational Number Systems

We denote the set $\{1, 2, 3, \dots\}$ of natural numbers by $\mathbb{N}$. The set $\mathbb{N}$ is defined by the following axioms:

- $1 \in \mathbb{N}$.
- If $n \in \mathbb{N}$, then $s(n) \in \mathbb{N}$ where $s(n) = n + 1$ is called the **successor** of n .
- 1 is not the successor of any element of $\mathbb{N}$.
- If $m, n \in \mathbb{N}$ and $s(m) = s(n)$, then $m = n$.
- Any subset of $\mathbb{N}$ which contains 1, and contains $s(n)$ whenever it contains n , must be equal to $\mathbb{N}$.

The axioms above are called the **Peano axioms** (or *Peano Postulates*).

Remark. Let $P(n)$ be a statement about $n \in \mathbb{N}$ and $S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\}$. Suppose $P(n)$ satisfies the following conditions:

- $P(1)$ is true.
- $P(n + 1)$ is true whenever $P(n)$ is true.

This implies that

- $1 \in S$.
- $n + 1 \in S$ whenever $n \in S$.

Thus $S = \mathbb{N}$, i.e., $P(n)$ is true for all $n \in \mathbb{N}$. This method of proof is called the **principle of mathematical induction**.

Theorem 1.1.1. Every nonempty subset of $\mathbb{N}$ has a least element, i.e., if $E \subseteq \mathbb{N}$ and $E \neq \emptyset$, then there exists an element $m \in E$ such that

$$m \leq n \text{ for all } n \in E.$$

This property is called the **well-ordering property** of $\mathbb{N}$.

Proof. Let S be a nonempty subset of $\mathbb{N}$ which does not have a least element. We define a set $T := \mathbb{N} \setminus S$, and $L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\}$. We will show that $L = \mathbb{N}$, which implies that $S = \emptyset$, a contradiction, by mathematical induction.

- **Base case:** $1 \in L$ since $\{1\} \subseteq T$, otherwise $1 \in S$ and 1 is the least element of S .
- **Inductive step:** Suppose $n \in L$, i.e., $\{1, 2, \dots, n\} \subseteq T$. Then $n + 1 \notin S$, otherwise $n + 1$ is the least element of S . Therefore, $n + 1 \in T$, and $\{1, 2, \dots, n + 1\} \subseteq T$, i.e., $n + 1 \in L$.

By the [principle of mathematical induction](#), $L = \mathbb{N}$. □

Remark. Now, we can define the set $\mathbb{Z}$ of integers and the set $\mathbb{Q}$ of rational numbers as follows:

- $\mathbb{Z} := \{m - n \mid m, n \in \mathbb{N}\}.$
- $\mathbb{Q} := \left\{ \frac{m}{n} \mid m \in \mathbb{Z}, n \in \mathbb{N} \right\}.$

Also, we denote the set of positive numbers or negative numbers as follows:

- $\mathbb{Z}^+ := \{n \mid n \in \mathbb{N}\}, \mathbb{Z}^- := \{-n \mid n \in \mathbb{N}\}.$
- $\mathbb{Q}^+ := \left\{ \frac{m}{n} \mid m \in \mathbb{N}, n \in \mathbb{N} \right\}, \mathbb{Q}^- := \left\{ \frac{m}{n} \mid m \in \mathbb{Z}^-, n \in \mathbb{N} \right\}.$

1.2 Ordered Sets

An **ordered set** is a set S together with a relation $<$ such that for any $x, y, z \in S$,

- (A) $x < y$ and $y < z$ implies $x < z$.
- (B) Only one of the relations $x < y$, $x = y$, or $x > y$ holds.

$x > y$ means $y < x$, $x \leq y$ means $(x < y \text{ or } x = y)$ and $x \geq y$ means $y \leq x$.

Remark. In an ordered set S , the relation $<$ is called a **order** (or a **total order**) on S and the set S is called **totally ordered**. If a relation $<$ satisfies (A) only, then the relation $<$ is called a **partial order** on S and the set S is called **partially ordered**.

Example 1.2.1. The set $\mathbb{N}$ and $\mathbb{Z}$ are ordered sets, where $m > n \iff m - n \in \mathbb{N}$. The set $\mathbb{Q}$ is also an ordered set, where $m > n \iff m - n \in \mathbb{Q}^+$.

Definition 1.2.2. Let S be an ordered set and $E \subseteq S$.

- E is **bounded above** if there exists $M \in S$ such that $x \leq M$ for all $x \in E$. Such an M is called an **upper bound** of E .
- E is **bounded below** if there exists $m \in S$ such that $m \leq x$ for all $x \in E$. Such an m is called a **lower bound** of E .

Theorem 1.2.3. Any nonempty subset of $\mathbb{Z}$ which is bounded below has a least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above has a greatest element.

Proof. Let $E \subseteq \mathbb{Z}$ be nonempty and bounded below. Then there exists $m \in \mathbb{Z}$ such that $m \leq x$ for all $x \in E$. Let $F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$. By [Theorem 1.1.1](#), F has a least element n .

$$n \leq x - m + 1 \text{ for all } x \in E \implies n + m - 1 \leq x \text{ for all } x \in E.$$

Since $n \in F$, there exists $x_0 \in E$ such that $n = x_0 - m + 1$, so $x_0 = n + m - 1$ is the least element of E .

Let $S \subseteq \mathbb{Z}$ be nonempty and bounded above. Then there exists $M \in \mathbb{Z}$ such that $x \leq M$ for all $x \in S$. Let $T := \{-x + M + 1 \mid x \in S\} \subseteq \mathbb{N}$. By [Theorem 1.1.1](#), T has a least element n .

$$n \leq -x + M + 1 \text{ for all } x \in S \implies M - n + 1 \geq x \text{ for all } x \in S.$$

Since $n \in T$, there exists $x_0 \in S$ such that $n = -x_0 + M + 1$, so $x_0 = M - n + 1$ is the greatest element of S . $\square$

Definition 1.2.4. Let S be an ordered set and $E \subseteq S$. Suppose E is bounded above. Suppose there exists an $\alpha \in S$ such that

- α is an upper bound of E .
- If $\beta < \alpha$, then β is not an upper bound of E .

Then α is called the *least upper bound* or **supremum** of E , denoted by

$$\alpha = \sup E.$$

Similarly, suppose E is bounded below. Suppose there exists a $\alpha \in S$ such that

- α is a lower bound of E .
- If $\beta > \alpha$, then β is not a lower bound of E .

Then α is called the *greatest lower bound* or **infimum** of E , denoted by

$$\alpha = \inf E.$$

Example 1.2.5. Let $S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$. For any $p \in S$, let $q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$. Then $q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0$, so $q \in S$. This implies that any $p \in S$ is not an upper bound of S .

For any $p \in \mathbb{Q}^+$ with $p^2 > 2$, p is an upper bound of S . Let $q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$. Then $q \in \mathbb{Q}^+$ and $q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0$, so q is an upper bound of S . This implies that such p is not the least upper bound of S .

Since there exists no $p \in \mathbb{Q}$ such that $p^2 = 2$, the set S has no least upper bound in $\mathbb{Q}^+$.

Definition 1.2.6. An ordered set S is said to have the **least upper bound property** if every nonempty subset $E \subseteq S$ that is bounded above has a least upper bound in S .

Theorem 1.2.7. If S is an ordered set with the least upper bound property, then any nonempty subset $E \subseteq S$ that is bounded below has a greatest lower bound in S .

Proof. Let E be a nonempty subset of S that is bounded below. Let L be the set of all lower bounds of E . Then L is nonempty and bounded above by any element of E . Let $\alpha = \sup L$. We claim that $\alpha = \inf E$.

Suppose $\alpha \notin L$. Then there exists $\beta \in E$ such that $\beta < \alpha$. Since $\alpha = \sup L$, there exists $\gamma \in L$ such that $\beta < \gamma \leq \alpha$. But γ is a lower bound of E , so $\gamma \leq \beta$, a contradiction. Therefore, $\alpha \in L$.

Since $\alpha = \sup L$, for any $\delta > \alpha$, $\delta \notin L$. Therefore, $\alpha = \inf E$. □

1.3 Fields and the Real Field

A set F is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$ for all $x, y \in F$.
- $x + y = y + x$ for all $x, y \in F$.
- $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- There exists an element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- For each $x \in F$, there exists an element $-x \in F$ such that $x + (-x) = 0$.

(B) Axioms for multiplication

- $x \cdot y \in F$ for all $x, y \in F$.
- $x \cdot y = y \cdot x$ for all $x, y \in F$.
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.
- There exists an element $1 \in F$ with $1 \neq 0$ such that $x \cdot 1 = x$ for all $x \in F$.
- For each $x \in F$ with $x \neq 0$, there exists an element $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$.

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in F$.

Remark. A set F is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If $x, y, z \in F$ and $x < y$, then $x + z < y + z$.
- If $x, y \in F$ and $x > 0$ and $y > 0$, then $x \cdot y > 0$.

Remark. There exists an ordered field $\mathbb{R}$ which has the [least upper bound property](#). Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield, i.e., $\mathbb{Q} \subseteq \mathbb{R}$ and the operations of addition and multiplication on $\mathbb{R}$ coincide with those on $\mathbb{Q}$.

We can construct $\mathbb{R}$ from $\mathbb{Q}$. But for now, we will assume the existence of $\mathbb{R}$ and its properties.

Theorem 1.3.1. *If $x, y \in \mathbb{R}$ and $x > 0$, then there exists some $n \in \mathbb{N}$ such that $nx > y$. This theorem is called the **Archimedean property** of $\mathbb{R}$.*

Proof. Suppose $x, y \in \mathbb{R}$ and $x > 0$. Let $E := \{nx \mid nx \leq y, n \in \mathbb{N}\}$. If $E = \emptyset$, which implies that $x > y$, $n = 1$ satisfies $nx > y$. Otherwise, E is nonempty and bounded above by y .

By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}$. Then $\alpha - x$ is not an upper bound of E . Therefore, there exists some $m \in \mathbb{N}$ such that $mx > \alpha - x \implies (m+1)x > \alpha$. Then $(m+1)x \notin E$, which implies that $(m+1)x > y$. Therefore, we can take $n = m+1$. $\square$

Remark. [Theorem 1.2.3](#) supposes that E is bounded in $\mathbb{Z}$. By [Theorem 1.3.1](#), we can always find an integer k such that $k < x$ or $x < k$ for any $x \in \mathbb{R}$. Therefore, [Theorem 1.2.3](#) also holds when E is bounded in $\mathbb{R}$, which is followed by the fact that E is also bounded in $\mathbb{Z}$.

Definition 1.3.2. For any $x \in \mathbb{R}$, the **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as follows:

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}, \quad \lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

whose existences are guaranteed by [Theorem 1.2.3](#).

Theorem 1.3.3. If $x, y \in \mathbb{R}$ and $x < y$, then there exists some $p \in \mathbb{Q}$ such that $x < p < y$. This theorem states that $\mathbb{Q}$ is **dense** in $\mathbb{R}$.

Proof. Let $\epsilon = y - x > 0$. By [Theorem 1.3.1](#), there exists some $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$. Let $S := \left\{k \in \mathbb{Z} \mid \frac{k}{n} < x\right\}$. Since $n\lfloor x \rfloor - 1 \in S$, S is nonempty and bounded above by nx . By [Theorem 1.2.3](#), S has a greatest element $m = \max S$.

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take $p = \frac{m+1}{n} \in \mathbb{Q}$. $\square$

Theorem 1.3.4. For each $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, there exists a unique $y \in \mathbb{R}^+$ such that $y^n = x$. We denote y by $\sqrt[n]{x}$ or $x^{\frac{1}{n}}$.

Proof. Let $E := \{t \in \mathbb{R}^+ \mid t^n < x\}$. Then $E \neq \emptyset$ since $2^{-1} \in E$ if $x \geq 1$, and $x^2 \in E$ otherwise. Moreover, E is bounded above by $\max\{x, 1\}$. By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}^+$. We will show that $\alpha^n = x$.

For any $a, b \in \mathbb{R}^+$ with $a < b$, we have

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a) n b^{n-1}.$$

Assume that $\alpha^n < x$. By [Theorem 1.3.3](#), we can choose $h \in \mathbb{R}^+$ such that

$$h < \min \left\{ 1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}} \right\}.$$

Put $a = \alpha$ and $b = \alpha + h$. Then we have

$$\begin{aligned} (\alpha + h)^n - \alpha^n &< hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha + h)^n &< \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

This shows that $\alpha + h \in E$, which contradicts the fact that $\alpha = \sup E$.

Assume that $\alpha^n > x$. By [Theorem 1.3.3](#), we can choose $h \in \mathbb{R}^+$ such that

$$h < \min \left\{ 1, \alpha, \frac{\alpha^n - x}{n(\alpha + 1)^{n-1}} \right\}.$$

Put $a = \alpha - h$ and $b = \alpha$. Then we have

$$\begin{aligned} \alpha^n - (\alpha - h)^n &< hn\alpha^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha - h)^n &> \alpha^n - hn(\alpha + 1)^{n-1} > \alpha^n - (\alpha^n - x) = x. \end{aligned}$$

This shows that $\alpha - h$ is an upper bound of E , which contradicts the fact that $\alpha = \sup E$.

Therefore, we have $\alpha^n = x$, which guarantees the *existence* of y . For any $\alpha, \beta \in \mathbb{R}^+$, $\alpha \neq \beta \implies \alpha^n \neq \beta^n$. Thus the *uniqueness* of y is guaranteed. $\square$

Theorem 1.3.5. Let $x \in \mathbb{R}$ and $b \in \mathbb{N}$ be given with $b \geq 2$. For any $n \in \mathbb{N}$, there exists a unique collection of integers $\{a_1, a_2, \dots, a_n\}$ such that

$$z_0 \leq x - \sum_{k=1}^n \frac{a_k}{b^k} < z_0 + \frac{1}{b^n}$$

where $z_0 = \lfloor x \rfloor$ and $a_k \in S = \{0, 1, 2, \dots, b-1\}$ for all $k = 1, 2, \dots, n$. The sequence $\{a_n\}_{n=1}^\infty$ is called the **base b expansion** of x . Especially, if $b = 10$, then the sequence is called the **decimal expansion** of x .

Proof. Let $T_n := \left\{ k \in S \mid z_n + \frac{k}{b} \leq b^n x \right\}$ for $n \in \mathbb{N} \cup \{0\}$, where $z_n = \lfloor b^n x \rfloor$. Since $0 \in T_n$ and T_n is bounded above, by [Theorem 1.2.3](#), T_n has a greatest element $a_n = \max T_{n-1}$ for each $n \in \mathbb{N}$.

First, we will show the *existence* by mathematical induction on n . Let P_n : The collection of integers $\{a_1, a_2, \dots, a_n\}$ satisfies

$$z_0 \leq x - \sum_{k=1}^n \frac{a_k}{b^k} < z_0 + \frac{1}{b^n}.$$

- **Base case:** Since $a_1 = \max T_0 \in T_0$, we have

$$z_0 + \frac{a_1}{b} \leq x < z_0 + \frac{a_1 + 1}{b} \implies z_0 \leq x - \frac{a_1}{b} < z_0 + \frac{1}{b}.$$

Thus P_1 is true.

- **Inductive step:** Assume that P_n is true for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned} z_0 &\leq x - \sum_{k=1}^n \frac{a_k}{b^k} < z_0 + \frac{1}{b^n} \\ \implies z_0 b^n &\leq b^n x - \sum_{k=1}^n a_k b^{n-k} < z_0 b^n + 1 \\ \implies z_0 b^n + \sum_{k=1}^n a_k b^{n-k} &\leq b^n x < z_0 b^n + \sum_{k=1}^n a_k b^{n-k} + 1. \end{aligned}$$

By [Definition 1.3.2](#), we have

$$z_n = z_0 b^n + \sum_{k=1}^n a_k b^{n-k}.$$

Since $a_{n+1} = \max T_n \in T_n$, we have

$$\begin{aligned} z_n + \frac{a_{n+1}}{b} &\leq b^n x < z_n + \frac{a_{n+1} + 1}{b} \\ \implies z_0 b^n + \sum_{k=1}^n a_k b^{n-k} + \frac{a_{n+1}}{b} &\leq b^n x < z_0 b^n + \sum_{k=1}^n a_k b^{n-k} + \frac{a_{n+1} + 1}{b} \\ \implies z_0 b^n + \sum_{k=1}^{n+1} a_k b^{n-k} &\leq b^n x < z_0 b^n + \sum_{k=1}^{n+1} a_k b^{n-k} + \frac{1}{b} \\ \implies z_0 + \sum_{k=1}^{n+1} \frac{a_k}{b^k} &\leq x < z_0 + \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \implies z_0 &\leq x - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < z_0 + \frac{1}{b^{n+1}}. \end{aligned}$$

Thus P_{n+1} is true whenever P_n is true.

By the [principle of mathematical induction](#), P_n is true for all $n \in \mathbb{N}$. Thus the *existence* of the collection $\{a_1, a_2, \dots, a_n\}$ for each $n \in \mathbb{N}$ is guaranteed.

Next, we will show the *uniqueness* of the sequence. Suppose that there exist two sequences of integers $\{a_1, a_2, \dots, a_n\}$ and $\{c_1, c_2, \dots, c_n\}$ such that

$$\begin{aligned} \bullet \quad & z_0 \leq x - \sum_{k=1}^n \frac{a_k}{b^k} < z_0 + \frac{1}{b^n}. \\ \bullet \quad & z_0 \leq x - \sum_{k=1}^n \frac{c_k}{b^k} < z_0 + \frac{1}{b^n}. \end{aligned}$$

for some $n \in \mathbb{N}$. Then we have both

$$\begin{aligned} \bullet \quad & \sum_{k=1}^n \frac{a_k}{b^k} \leq x - z_0 < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}. \\ \bullet \quad & \sum_{k=1}^n \frac{c_k}{b^k} \leq x - z_0 < \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n}. \end{aligned}$$

By combining the two inequalities, we have

$$\begin{aligned} \sum_{k=1}^n \left| \frac{a_k - c_k}{b^k} \right| < \frac{1}{b^n} &\implies \left| \frac{a_k - c_k}{b^k} \right| < \frac{1}{b^n} \text{ for all } k = 1, 2, \dots, n \\ &\implies |a_k - c_k| < b^{k-n} \leq 1 \text{ for all } k = 1, 2, \dots, n. \end{aligned}$$

Since $a_k, c_k \in S$ and so is $|a_k - c_k|$, we have $a_k = c_k$ for all $k = 1, 2, \dots, n$. Thus the *uniqueness* of the collection $\{a_1, a_2, \dots, a_n\}$ for each $n \in \mathbb{N}$ is guaranteed. $\square$

Theorem 1.3.6. *Let $\{a_n\}_{n=1}^\infty$ be the base b expansion of $a \in \mathbb{R}$, and $\{c_n\}_{n=1}^\infty$ be the base b expansion of $c \in \mathbb{R}$ for some $b \in \mathbb{N}$ with $b \geq 2$. If $a_n = c_n$ for all $n \in \mathbb{N}$, then $a = c$.*

Proof. We will prove by contraposition. Suppose that $a \neq c$. Without loss of generality, assume that $a < c$. Then we have $c - a > 0$. By [Theorem 1.3.1](#), there exists some $n \in \mathbb{N}$ such that

$$b^n > \frac{1}{c - a} \implies \frac{1}{b^n} < c - a.$$

Let $z_0 = \lfloor a \rfloor$ and $w_0 = \lfloor c \rfloor$. Then we have $z_0 \leq a < z_0 + 1$ and $w_0 \leq c < w_0 + 1$. If $z_0 \neq w_0$, then $a_1 \neq c_1$, which contradicts the assumption. $\square$

1.4 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on $\overline{\mathbb{R}}$:

- $x + (+\infty) = +\infty$ for all $x \in \mathbb{R} \cup \{\infty\}$.
- $x + (-\infty) = -\infty$ for all $x \in \mathbb{R} \cup \{-\infty\}$.
- $x \cdot (\pm\infty) = \pm\infty$ for all $x \in \mathbb{R}^+ \cup \{\infty\}$.
- $x \cdot (\pm\infty) = \mp\infty$ for all $x \in \mathbb{R}^- \cup \{-\infty\}$.
- $\frac{x}{\pm\infty} = 0$ for all $x \in \mathbb{R}$.

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

Theorem 1.4.1. *Every subset of $\overline{\mathbb{R}}$ has a least upper bound and a greatest lower bound in $\overline{\mathbb{R}}$.*

Proof. Let $E \subseteq \overline{\mathbb{R}}$. If $E = \emptyset$, then $\sup E = -\infty$ and $\inf E = +\infty$. If $E \neq \emptyset$, then we have the following cases:

- If E is bounded above in $\mathbb{R}$, then $\sup E$ exists in $\mathbb{R}$ by the least upper bound property of $\mathbb{R}$.
- If E is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$.
- If E is bounded below in $\mathbb{R}$, then $\inf E$ exists in $\mathbb{R}$ by the greatest lower bound property of $\mathbb{R}$.
- If E is not bounded below in $\mathbb{R}$, then $\inf E = -\infty$.

□

1.5 The Complex Field

The complex field $\mathbb{C}$ is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on $\mathbb{C}$:

- $(x, y) + (u, v) = (x + u, y + v)$ for all $(x, y), (u, v) \in \mathbb{C}$.
- $(x, y) \cdot (u, v) = (xu - yv, xv + yu)$ for all $(x, y), (u, v) \in \mathbb{C}$.

$\mathbb{C}$ is a field with the following properties:

- The additive identity is $(0, 0)$ and the multiplicative identity is $(1, 0)$.
- The additive inverse of (x, y) is $(-x, -y)$ and the multiplicative inverse of (x, y) is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

for all $(x, y) \in \mathbb{C}$ with $(x, y) \neq (0, 0)$.

Remark. The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \rightarrow \mathbb{R}$$

defined by $f((x, 0)) = x$ is an *isomorphism*. Then we can identify $\mathbb{R}$ with the subfield $\{(x, 0) \mid x \in \mathbb{R}\}$ of $\mathbb{C}$. Now we will denote $(x, 0) \in \mathbb{C}$ by $x \in \mathbb{R}$.

Remark. We define $i = (0, 1) \in \mathbb{C}$. Then we have $i^2 = -1$, and any complex number $(x, y) \in \mathbb{C}$ can be written as

$$(x, y) = x + yi.$$

Definition 1.5.1. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. The **real part** of z is defined as

$$\Re(z) = x \text{ or } \operatorname{Re}(z) = x,$$

the **imaginary part** of z is defined as

$$\Im(z) = y \text{ or } \operatorname{Im}(z) = y,$$

and the **complex conjugate** of z is defined as

$$\bar{z} = x - yi.$$

Theorem 1.5.2. Let $z, w \in \mathbb{C}$. Then we have the following properties:

- $\overline{(\bar{z})} = z$.
- $z = \bar{z} \iff z \in \mathbb{R}$.
- $\overline{z + w} = \bar{z} + \bar{w}$.

- $\overline{zw} = \overline{z} \cdot \overline{w}$.
- $z + \overline{z} = 2\operatorname{Re}(z)$ and $z - \overline{z} = 2i\operatorname{Im}(z)$.
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$.

Proof. The proof follows directly from the definitions and properties of the operations on $\mathbb{C}$. $\square$

Definition 1.5.3. Let $z \in \mathbb{C}$. the **modulus** of z is defined as

$$|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The *existence* and *uniqueness* of $|z|$ follows from the last part of [Theorem 1.5.2](#) and [Theorem 1.3.4](#).

Theorem 1.5.4. Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ and $b_1, b_2, \dots, b_n \in \mathbb{C}$. Then we have

$$\left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2.$$

This theorem is called the **Cauchy-Schwarz inequality**.

Proof. Let $i, j \in \mathbb{N}$ with $1 \leq i < j \leq n$. Then we have

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned} \sum_{i < j} (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) &= \sum_{i \neq j} (a_i \overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^n a_i \overline{b_i} \cdot \sum_{j=1}^n \overline{a_j} b_j - \sum_{k=1}^n a_k \overline{b_k} \cdot \overline{a_k} b_k \\ &= \left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

Therefore, we have

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \geq 0.$$

□

Corollary 1.5.5. *Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ and $b_1, b_2, \dots, b_n \in \mathbb{R}$. Then we have*

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2.$$

Theorem 1.5.6. *Let $z, w \in \mathbb{C}$. Then we have the following properties:*

(A) $|zw| = |z| \cdot |w|.$

(B) $|z + w| \leq |z| + |w|.$

(C) $|z| - |w| \leq |z - w|.$

Proof. Let $z = x + yi$ and $w = u + vi$ for some $x, y, u, v \in \mathbb{R}$. Then we have

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2.$

(B) By [Corollary 1.5.5](#), we have $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2).$

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2. \end{aligned}$$

(C) Again, by [Corollary 1.5.5](#), we have

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2. \end{aligned}$$

□

1.6 Euclidean Spaces

For $n \in \mathbb{N}$, we define the **Euclidean n -space** $\mathbb{R}^n$ as the set of all ordered n -tuples of real numbers, i.e.,

$$\mathbb{R}^n := \{ \mathbf{x} = (x_1, x_2, \dots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \dots, n \}$$

where $x_1, x_2, \dots, x_n$ are called the **coordinates** of $\mathbf{x}$. The elements of $\mathbb{R}^n$ with $n > 1$ are called **vectors** or *points*, which are denoted by boldface letters.

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. We define the following operations on $\mathbb{R}^n$:

- $\mathbf{x} + \mathbf{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$.
- $\alpha \mathbf{x} = (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$.

Then $\mathbb{R}^n$ is a *vector space* over $\mathbb{R}$ with the zero vector $\mathbf{0} = (0, 0, \dots, 0)$.

Definition 1.6.1. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The **inner product** of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\mathbf{x} \cdot \mathbf{y} = \sum_{k=1}^n x_k y_k.$$

The **norm** of $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}.$$

The **distance** between $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}.$$

We can confirm that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.6.2. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. Then we have the following properties:

- (A) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$.
- (B) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.
- (C) $\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$.

Proof. (A) By [Corollary 1.5.5](#), we have

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2.$$

(B) By part (A), we have

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

(C) By part (B), we have

$$\begin{aligned} \|\mathbf{x}\| &= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|. \\ \|\mathbf{y}\| &= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| \implies \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

□

Chapter 2

Basic Topology

2.1 Relations

A **relation** $\sim$ on a set S defines a subset R of $S \times S$, and vice versa.

$$R := \{(x, y) \in S \times S \mid x \sim y\}, \quad x \sim y \iff (x, y) \in R.$$

Definition 2.1.1. A relation $\sim$ on a set S is called an **equivalence relation** if it satisfies the following properties:

- Reflexivity: $x \sim x$ for all $x \in S$.
- Symmetry: If $x \sim y$, then $y \sim x$ for all $x, y \in S$.
- Transitivity: If $x \sim y$ and $y \sim z$, then $x \sim z$ for all $x, y, z \in S$.

Definition 2.1.2. A **partition** of a set S is a collection $\{A_\alpha\}_{\alpha \in I}$ of nonempty subsets of S such that

- $A_\alpha \cap A_\beta = \emptyset$ for all $\alpha, \beta \in I$ with $\alpha \neq \beta$.
- $\bigcup_{\alpha \in I} A_\alpha = S$.

Definition 2.1.3. Let $\sim$ be an equivalence relation on a set S . For each $x \in S$, the **equivalence class** of x is defined as

$$[x] := \{y \in S \mid y \sim x\}.$$

We denote the set of all equivalence classes of S by $S/\sim := \{[x] \mid x \in S\}$.

Theorem 2.1.4. *Let $\sim$ be an equivalence relation on a set S . For any $x, y \in S$, we have either*

$$[x] = [y] \quad \text{or} \quad [x] \cap [y] = \emptyset.$$

Therefore, $S/\sim$ is a partition of S .

Proof. Suppose $[x] \cap [y] \neq \emptyset$. Then there exists $z \in S$ such that $z \in [x]$ and $z \in [y]$. This implies that $z \sim x$ and $z \sim y$. By the symmetry and transitivity of $\sim$, we have $x \sim z$ and $z \sim y$, which implies that $x \sim y$.

Therefore, for any $w \in [x]$, we have $w \sim x$ and $x \sim y$, which implies that $w \sim y$, so $w \in [y]$. This shows that $[x] \subseteq [y]$. Similarly, we can show that $[y] \subseteq [x]$, so $[x] = [y]$. $\square$

Theorem 2.1.5. *Let $\{A_\alpha\}_{\alpha \in I}$ be a partition of a set S . Define a relation $\sim$ on S by*

$$x \sim y \iff x, y \in A_\alpha \text{ for some } \alpha \in I.$$

Then $\sim$ is an equivalence relation on S , and the equivalence classes of $\sim$ are precisely the sets A_α .

Proof. We need to show that $\sim$ is reflexive, symmetric, and transitive.

- Reflexivity: For any $x \in S$, there exists $\alpha \in I$ such that $x \in A_\alpha$, so $x \sim x$.
- Symmetry: If $x \sim y$, then there exists $\alpha \in I$ such that $x, y \in A_\alpha$, so $y \sim x$.
- Transitivity: If $x \sim y$ and $y \sim z$, then there exist $\alpha, \beta \in I$ such that $x, y \in A_\alpha$ and $y, z \in A_\beta$. Since $y \in A_\alpha \cap A_\beta$ and the sets $\{A_\alpha\}_{\alpha \in I}$ are disjoint, we must have $\alpha = \beta$, so $x, z \in A_\alpha$ and $x \sim z$.

Therefore, $\sim$ is an equivalence relation on S . The equivalence class of any $x \in S$ is precisely the set A_α containing x , so the equivalence classes of $\sim$ are exactly the sets A_α . $\square$

2.2 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain** X .
- A **codomain** Y .
- A **rule** f which associates each element $x \in X$ to a *unique* element $y \in Y$.

f is said to be a *function* of X into Y , which is denoted by $f : X \rightarrow Y$. For each $x \in X$, the *unique* element $y \in Y$ associated with x is denoted by $f(x)$.

Remark. Let $f : X \rightarrow Y$ be a function. For any subset $E \subseteq X$, we define the **image** of E under f as

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f .

Remark. Let $f : X \rightarrow Y$ be a function. For any subset $E \subseteq Y$, we define the **inverse image** of E under f as

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X.$$

Definition 2.2.1. Let $f : X \rightarrow Y$ be a function. We define the properties of f as follows:

- f is **injective** (or *one-to-one*) if for any $x_1, x_2 \in X$,

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

- f is **surjective** (or *onto*) if $f(X) = Y$.
- f is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

Definition 2.2.2. Let $f : X \rightarrow Y$ be a bijective function. Then for any $y \in Y$, there exists a *unique* $x \in X$ such that $f(x) = y$. Therefore, we can define a function $f^{-1} : Y \rightarrow X$ by $f^{-1}(y) = x$. f^{-1} is called the **inverse function** of f .

Theorem 2.2.3. Let $f : X \rightarrow Y$ be a bijective function. Then the inverse function $f^{-1} : Y \rightarrow X$ is bijective.

Proof. Let $y_1, y_2 \in Y$. Suppose that $f^{-1}(y_1) = f^{-1}(y_2) = x$. Then, by [Definition 2.2.2](#), we have both $f(x) = y_1$ and $f(x) = y_2$. Since f is a function, we have $y_1 = y_2$, which implies that f^{-1} is injective.

By the definition of inverse image, we have $f^{-1}(Y) \subseteq X$. Also, for any $x \in X$, we have $f(x) \in Y$, which implies that $x \in f^{-1}(Y)$. Therefore, we have $f^{-1}(Y) \supseteq X \implies f^{-1}(Y) = X$, which implies that f^{-1} is surjective. $\square$

Definition 2.2.4. Let $f : X \rightarrow Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ such that $f(E) \subseteq F$. Then we define the restriction functions as follows:

- The **restriction** of f to E is defined by

$$f|_E : E \rightarrow Y, \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$

- The **Corestriction** of f to F is defined by

$$f|_F : X \rightarrow F, \quad f|_F(x) = f(x) \quad \text{for all } x \in X.$$

- The **Bi-restriction** of f to E and F is defined by

$$f|_E^F : E \rightarrow F, \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

Theorem 2.2.5. *Let $f : X \rightarrow Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ such that $f(E) \subseteq F$. Then we have the following properties:*

- (A) *If f is injective, then $f|_E^F$ is also injective.*
- (B) *If $f(E) = F$, then $f|_E^F$ is surjective.*
- (C) *If f is injective and $f(E) = F$, then $f|_E^F$ is bijective.*

Proof. (C) follows from (A) and (B). Let $x_1, x_2 \in E$.

- (A) Assume that f is injective. Then we have

$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

- (B) Assume that $f(E) = F$. Then for any $y \in F$, there exists $x \in E$ such that $f(x) = y$. Therefore, we have $f|_E^F(E) = F$.

□

Definition 2.2.6. Let X, Y, Z be sets and $f : X \rightarrow Y, g : Y \rightarrow Z$ be functions. Then we define the **composition** of f and g as the function $g \circ f : X \rightarrow Z$ defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

Theorem 2.2.7. *Let X, Y, Z be sets and $f : X \rightarrow Y, g : Y \rightarrow Z$ be functions. Then we have the following properties:*

- (A) *If f and g are injective, then $g \circ f$ is also injective.*

(B) If f and g are surjective, then $g \circ f$ is also surjective.

(C) If f and g are bijective, then $g \circ f$ is also bijective.

Proof. (C) follows from (A) and (B). Let $x_1, x_2 \in X$.

(A) Assume that f and g are injective. Then we have

$$\begin{aligned} (g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ &\implies f(x_1) = f(x_2) \implies x_1 = x_2. \end{aligned}$$

(B) Assume that f and g are surjective. Then for any $z \in Z$, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, we have $(g \circ f)(x) = g(f(x)) = g(y) = z$.

□

Theorem 2.2.8. Let S be a set of sets. For [Sections 2.3](#) and [2.4](#), we consider the relation $\sim$ defined on S by

$$X \sim Y \iff \text{There exists a bijection } f : X \rightarrow Y.$$

The relation $\sim$ is an equivalence relation on S .

Proof. Let $X, Y, Z \in S$.

- **Reflexivity:** The identity function $id_X : X \rightarrow X$ defined by $id_X(x) = x$ for all $x \in X$ is bijective. Therefore, we have $X \sim X$.
- **Symmetry:** Suppose that $X \sim Y$. Then there exists a bijection $f : X \rightarrow Y$. By [Theorem 2.2.3](#), the inverse function $f^{-1} : Y \rightarrow X$ is also bijective. Therefore, we have $Y \sim X$.
- **Transitivity:** Suppose that $X \sim Y$ and $Y \sim Z$. Then there exist bijections $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. By [Theorem 2.2.7](#), the composition $g \circ f : X \rightarrow Z$ is also bijective. Therefore, we have $X \sim Z$.

□

2.3 Finite Sets

Let $J_n := \{1, 2, \dots, n\}$ for each $n \in \mathbb{N}$. A set X is said to be **finite** if $J_n \sim X$ for some $n \in \mathbb{N}$. In this case, n is called the **cardinality** of X , denoted by $|X|$. Especially, if $X = \emptyset$, we define $|X| = 0$. X is said to be **infinite** if X is not finite.

Lemma 2.3.1. Every subset of J_n is finite for each $n \in \mathbb{N}$.

Proof. We will prove the lemma by mathematical induction on n . Let $P(n)$: Every subset of J_n is finite.

- **Base case:** Let $n = 1$. Then $J_1 = \{1\}$, whose only subsets are $\emptyset$ and J_1 . $J_1 \sim J_1$ by $f := id_{J_1}$, and $\emptyset$ is finite by definition. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $A \subseteq J_{n+1}$. We have two cases:
 - If $n + 1 \notin A$, then $A \subseteq J_n$, which is finite by the inductive hypothesis.
 - If $n + 1 \in A$, then $A = B \cup \{n + 1\}$ for some $B \subseteq J_n$. By the inductive hypothesis, B is finite, so there exists a bijection $f : J_m \rightarrow B$ for some $m \in \mathbb{N}$. Let $g : J_{m+1} \rightarrow A$ be defined by

$$g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n + 1 & \text{if } k = m + 1 \end{cases}.$$

Then g is bijective, which implies that A is finite.

Thus $P(n + 1)$ is true whenever $P(n)$ is true.

By the [principle of mathematical induction](#), $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.3.2. *Every subset of a finite set is finite.*

Proof. Let X be a finite set and $Y \subseteq X$. Then there exists a bijection $f : J_n \rightarrow X$ for some $n \in \mathbb{N}$. Let $Z = f^{-1}(Y) \subseteq J_n$. Since $Z \subseteq J_n$, Z is finite by [Lemma 2.3.1](#), so there exists a bijection $g : J_m \rightarrow Z$ for some $m \in \mathbb{N}$.

By [Theorem 2.2.5](#), $f|_Z^Y$ is bijective. Then, by [Theorem 2.2.7](#), the composition $f|_Z^Y \circ g : J_m \rightarrow Y$ is bijective. Thus Y is finite. □

Theorem 2.3.3. *Let X be a finite ordered set. Then X has both a least element and a greatest element.*

Proof. Since X is finite, there exists a bijection $f : J_n \rightarrow X$ for some $n \in \mathbb{N}$. The elements of X can be listed as $X = \{a_1, a_2, \dots, a_n\}$, where $a_k = f(k)$ for each $k = 1, 2, \dots, n$.

First, we will show that X has a least element by mathematical induction on n . Let $P(n)$: Any finite ordered set of cardinality n has a least element.

- **Base case:** Let $n = 1$. Then $X = \{a_1\}$, which has a least element a_1 .

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $X = \{a_1, a_2, \dots, a_{n+1}\}$. By the inductive hypothesis, the subset $\{a_1, a_2, \dots, a_n\}$ has a least element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:
 - If $a_k \leq a_{n+1}$, then a_k is the least element of X .
 - If $a_{n+1} < a_k$, then a_{n+1} is the least element of X .

Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the [principle of mathematical induction](#), $P(n)$ is true for all $n \in \mathbb{N}$.

Next, we will show that X has a greatest element by mathematical induction on n . Let $Q(n)$: Any finite ordered set of cardinality n has a greatest element.

- **Base case:** Let $n = 1$. Then $X = \{a_1\}$, which has a greatest element a_1 .
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. Let $X = \{a_1, a_2, \dots, a_{n+1}\}$. By the inductive hypothesis, the subset $\{a_1, a_2, \dots, a_n\}$ has a greatest element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:
 - If $a_k \geq a_{n+1}$, then a_k is the greatest element of X .
 - If $a_{n+1} > a_k$, then a_{n+1} is the greatest element of X .

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the [principle of mathematical induction](#), $Q(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.3.4. Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:

- (A) If there exists an injection $f : J_m \rightarrow J_n$, then $m \leq n$.
- (B) If there exists a surjection $f : J_m \rightarrow J_n$, then $m \geq n$.
- (C) If there exists a bijection $f : J_m \rightarrow J_n$, then $m = n$.

This lemma guarantees the uniqueness of the cardinality of a finite set. If $J_m \sim X$ and $J_n \sim X$, then $J_m \sim J_n$, which implies that $m = n$ by (C).

Proof. We will prove (A) by mathematical induction on m , (B) by mathematical induction on n , and (C) follows from (A) and (B).

- (A) Let $n \in \mathbb{N}$ be fixed. Let $P(m)$: If there exists an injective function $f_m : J_m \rightarrow J_n$, then $m \leq n$.
 - **Base case:** Let $m = 1$. Then $m \leq n$ holds.

- **Inductive step:** Assume that $P(m)$ is true for some $m \in \mathbb{N}$. If $n = 1$, there does not exist an injective function $f_{m+1} : J_{m+1} \rightarrow J_n$ since $f_{m+1}(1) \neq f_{m+1}(m+1)$, where $P(m+1)$ is true.

Otherwise, suppose that such a function f_{m+1} exists. We have two cases:

- $f_{m+1}(m+1) = n$. Let $f_m \equiv f_{m+1}|_{J_m}^{J_{n-1}}$. Then $f_m : J_m \rightarrow J_{n-1}$ is injective by [Theorem 2.2.5](#). By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.
- $f_{m+1}(m+1) = x \neq n$. Let $g_n : J_n \rightarrow J_n$ be defined by

$$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}.$$

Then g_n is bijective. Let $f_m \equiv (g_n \circ f_{m+1})|_{J_m}^{J_{n-1}}$. Then $f_m : J_m \rightarrow J_{n-1}$ is injective by [Theorems 2.2.5](#) and [2.2.7](#). By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

Thus $P(m+1)$ is true whenever $P(m)$ is true.

By the [principle of mathematical induction](#), $P(m)$ is true for all $m \in \mathbb{N}$.

- (B) Let $m \in \mathbb{N}$ be fixed. Let $Q(n)$: If there exists a surjective function $f_n : J_m \rightarrow J_n$, then $m \geq n$.

- **Base case:** Let $n = 1$. Then $m \geq n$ holds.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. If $m = 1$, there does not exist a surjective function $f_{n+1} : J_m \rightarrow J_{n+1}$ since $f_{n+1}(1)$ is unique, where $Q(n+1)$ is true.

Otherwise, suppose that such a function f_{n+1} exists. We have two cases:

- $f_{n+1}(m) = n+1$. Let $K_{n+1} := (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m$. Let $f_n : J_m \rightarrow J_n$ be defined by

$$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}.$$

Then f_n is surjective. By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

- $f_{n+1}(m) = x \neq n+1$. Let $g_{n+1} : J_{n+1} \rightarrow J_{n+1}$ be defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}.$$

Then g_{n+1} is bijective. Let $K_{n+1} := (g_{n+1})^{-1}(\{n+1\}) \subseteq J_m$. Let $f_n : J_m \rightarrow J_n$ be defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}.$$

Then f_n is surjective by [Theorem 2.2.7](#). By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the [principle of mathematical induction](#), $Q(n)$ is true for all $n \in \mathbb{N}$.

□

Lemma 2.3.5. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

- (A) *If $m \leq n$, then there exists an injection $f : J_m \rightarrow J_n$.*
- (B) *If $m \geq n$, then there exists a surjection $f : J_m \rightarrow J_n$.*
- (C) *If $m = n$, then there exists a bijection $f : J_m \rightarrow J_n$.*

This lemma is the converse of [Lemma 2.3.4](#).

Proof. We will prove the lemma by explicitly constructing the functions.

- (A) If $m \leq n$, define $f : J_m \rightarrow J_n$ by $f(i) = i$ for all $i \in J_m$. Then f is an injection.
- (B) If $m \geq n$, define $f : J_m \rightarrow J_n$ by $f(i) = i$ for all $i \in J_n \subseteq J_m$ and $f(i) = n$ for all $i \in J_m \setminus J_n$. Then f is a surjection.
- (C) If $m = n$, define $f : J_m \rightarrow J_n$ by $f(i) = i$ for all $i \in J_m$. Then f is a bijection.

□

Theorem 2.3.6. *Let X, Y be finite sets. Then we have the following properties:*

- (A) *If there exists an injection $f : X \rightarrow Y$, then $|X| \leq |Y|$.*
- (B) *If there exists a surjection $f : X \rightarrow Y$, then $|X| \geq |Y|$.*
- (C) *If there exists a bijection $f : X \rightarrow Y$, then $|X| = |Y|$.*

Moreover, the converses of (A), (B), and (C) are also true.

The contraposition of (A) is called the **pigeonhole principle**, which states that if $|X| > |Y|$, then there exists no injection $f : X \rightarrow Y$.

Proof. Let $|X| = m$ and $|Y| = n$ for some $m, n \in \mathbb{N}$. Then there exist bijections $f_X : J_m \rightarrow X$ and $f_Y : J_n \rightarrow Y$. By Theorems 2.2.3 and 2.2.7, the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \rightarrow J_n$$

is injective, surjective, or bijective if f is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.4, we have the desired results:

- (A) If f is injective, then F is injective, which implies that $m \leq n$.
- (B) If f is surjective, then F is surjective, which implies that $m \geq n$.
- (C) If f is bijective, then F is bijective, which implies that $m = n$.

Also, by Theorems 2.2.3 and 2.2.7, the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \rightarrow Y$$

is injective, surjective, or bijective if $G : J_m \rightarrow J_n$ is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.5, we have the desired converses:

- (A) If $m \leq n$, then there exists an injective function $G : J_m \rightarrow J_n$, which implies that there exists an injective function $f : X \rightarrow Y$.
- (B) If $m \geq n$, then there exists a surjective function $G : J_m \rightarrow J_n$, which implies that there exists a surjective function $f : X \rightarrow Y$.
- (C) If $m = n$, then there exists a bijective function $G : J_m \rightarrow J_n$, which implies that there exists a bijective function $f : X \rightarrow Y$.

□

Lemma 2.3.7. Let X, Y be finite sets with $X \cap Y = \emptyset$. Then the union $X \cup Y$ is finite.

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \rightarrow X$ and $g : J_n \rightarrow Y$. Let $h : J_{m+n} \rightarrow X \cup Y$ be defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i - m), & \text{if } m < i \leq m + n \end{cases}.$$

Then h is bijective, which implies that $X \cup Y$ is finite.

□

Theorem 2.3.8. *Let X, Y be finite sets. Then the union $X \cup Y$ and intersection $X \cap Y$ are finite.*

Proof. Let A, B, C be defined by

- $A = X \setminus Y \subseteq X$.
- $B = Y \setminus X \subseteq Y$.
- $C = X \cap Y \subseteq X$.

Then A, B, C are finite by [Theorem 2.3.2](#) and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$.
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$.
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$.

By [Lemma 2.3.7](#), $X, Y, X \cup Y$ are finite. □

Theorem 2.3.9. *Let $X_1, X_2, \dots, X_n$ be finite sets for some $n \in \mathbb{N}$ with $n \geq 2$. Then the union $\bigcup_{k=1}^n X_k$ and intersection $\bigcap_{k=1}^n X_k$ are finite.*

Proof. Since $\bigcap_{k=1}^n X_k \subseteq X_1$, $\bigcap_{k=1}^n X_k$ is finite by [Theorem 2.3.2](#) for any $n \in \mathbb{N}$. Let

$P(n) : \bigcup_{k=1}^n X_k$ is finite for some $n \in \mathbb{N}$, given that $X_1, X_2, \dots, X_n$ are finite sets.

- **Base case:** Let X_1, X_2 be finite sets. Then, by [Theorem 2.3.8](#), $\bigcup_{k=1}^2 X_k = X_1 \cup X_2$ is finite. Thus, $P(2)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be finite sets. Then we have

$$\bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^n X_k \right) \cup X_{n+1}.$$

By the inductive hypothesis, $\bigcup_{k=1}^n X_k$ is finite. Then, by [Theorem 2.3.8](#), $\bigcup_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the [principle of mathematical induction](#), $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.3.10. *Let X be a nonempty set. Then there exists no surjection $f : X \rightarrow \mathcal{P}(X)$, where*

$$\mathcal{P}(X) = \{E \mid E \subseteq X\}$$

*is the **power set** of X . In particular, if X is finite, then $|\mathcal{P}(X)| > |X|$. This theorem is called **Cantor's theorem**.*

Proof. Let $S = \{x \in X \mid x \notin f(x)\}$. Then $S \subseteq X$, which implies that $S \in \mathcal{P}(X)$. Suppose that there exists a surjection $f : X \rightarrow \mathcal{P}(X)$. Then there exists $x \in X$ such that $f(x) = S$.

If $x \in S$, then $x \notin f(x) = S$, which is a contradiction. If $x \notin S$, then $x \in f(x) = S$, which is also a contradiction. Therefore, there exists no surjection $f : X \rightarrow \mathcal{P}(X)$.

If X is finite, by [Theorem 2.3.6](#), there exists no surjection $f : X \rightarrow \mathcal{P}(X)$ if and only if $|X| < |\mathcal{P}(X)|$. $\square$

2.4 Countable and Uncountable Sets

A set X is said to be **countable** if $\mathbb{N} \sim X$. A set X is said to be **uncountable** if it is neither finite nor countable. A set X is said to be **at most countable** if X is either finite or countable.

Theorem 2.4.1. *Let X be an infinite set and Y be a finite set. Then $X \setminus Y$ is infinite.*

Proof. Suppose $X \setminus Y$ is finite. Since Y is finite, by [Theorem 2.3.8](#), $(X \setminus Y) \cup Y$ is finite. Then, by [Theorem 2.3.2](#), $X \subseteq (X \setminus Y) \cup Y$ is finite, which contradicts the assumption. Thus, $X \setminus Y$ is infinite. $\square$

Theorem 2.4.2. *Let X be an infinite set. If $X \sim Y$, then Y is infinite.*

Proof. Suppose Y is finite. Then there exists a bijection $f : J_n \rightarrow Y$ for some $n \in \mathbb{N}$. Since $X \sim Y$, there exists a bijection $g : X \rightarrow Y$.

Then the composition $g^{-1} \circ f : J_n \rightarrow X$ is a bijection, which implies that X is finite. This contradicts the assumption that X is infinite. Thus, Y is infinite. $\square$

Lemma 2.4.3. $\mathbb{N}$ is infinite.

Proof. Suppose $\mathbb{N}$ is finite. Then there exists a bijection $f : J_n \rightarrow \mathbb{N}$ for some $n \in \mathbb{N}$. Then, by [Theorem 2.3.2](#), $f(J_n)$ is finite. By [Theorem 2.3.3](#), $f(J_n)$ has a greatest element, say m .

Then we have $m+1 \in \mathbb{N}$ and $m < m+1 \notin f(J_n)$, which contradicts the assumption that f is surjective. Thus, $\mathbb{N}$ is infinite. $\square$

Theorem 2.4.4. *Every countable set is infinite.*

Proof. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \rightarrow X$. By [Theorem 2.4.2](#) and [lemma 2.4.3](#), X is infinite. $\square$

Remark. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \rightarrow X$. The elements of X can be listed as $X = \{x_1, x_2, x_3, \dots\}$, where $x_n = f(n)$ for all $n \in \mathbb{N}$. In other words, the elements of X can be arranged in a sequence.

Theorem 2.4.5. *$\mathbb{Z}$ is countable.*

Proof. Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be defined by

$$f(n) = \begin{cases} -\frac{n}{2}, & \text{if } n \text{ is even} \\ \frac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}.$$

We will show that f is bijective, which implies that $\mathbb{Z}$ is countable.

- **Injectivity:** Suppose $f(n_1) = f(n_2)$ for some $n_1, n_2 \in \mathbb{N}$. Since $f(n) \geq 0$ for odd n and $f(n) < 0$ for even n , we have n_1 and n_2 are either both even or both odd.

- If both n_1 and n_2 are even, then $-\frac{n_1}{2} = -\frac{n_2}{2}$, which implies $n_1 = n_2$.
- If both n_1 and n_2 are odd, then $\frac{n_1-1}{2} = \frac{n_2-1}{2}$, which implies $n_1 = n_2$.

Thus, f is injective.

- **Surjectivity:** Let $k \in \mathbb{Z}$. We need to find an $n \in \mathbb{N}$ such that $f(n) = k$.

- If $k \geq 0$, let $n = 2k + 1$. Then $f(n) = \frac{n-1}{2} = \frac{2k}{2} = k$.
- If $k < 0$, let $n = -2k$. Then $f(n) = -\frac{n}{2} = -\frac{-2k}{2} = k$.

Thus, f is surjective.

Since f is both injective and surjective, we have f is bijective. $\square$

Lemma 2.4.6. *Every infinite subset of $\mathbb{N}$ is countable.*

Proof. Let $X \subseteq \mathbb{N}$ be an infinite subset. Let $X_1 := X$ and define a function $f : \mathbb{N} \rightarrow X$ as follows:

- $f(n) = \min X_n$
- $X_{n+1} := X_n \setminus \{f(n)\}$

for each $n \in \mathbb{N}$. By [Theorem 2.4.1](#), X_n is infinite for each $n \in \mathbb{N}$, so is nonempty. Then, by [Theorem 1.1.1](#), X_n has a least element, so f is well-defined. We will show that f is bijective, which implies that X is countable.

- **Injectivity:** Suppose $n_1 \neq n_2$ for some $n_1, n_2 \in \mathbb{N}$. Without loss of generality, assume that $n_1 < n_2$. Then $f(n_1) \in X_{n_1}$ and $f(n_2) \in X_{n_2}$. Since $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$, we have $f(n_2) \neq f(n_1)$. Thus, f is injective.
- **Surjectivity:** Let $x \in X$. Let $S := \{n \in \mathbb{N} \mid n < x\}$. If $S = \emptyset$, then $f(1) = x$. Otherwise, by [Lemma 2.3.1](#), $S \subseteq \{1, 2, \dots, x-1\}$ is finite. Let $k = |S|$, $S = \{s_1, s_2, \dots, s_k\}$ with $s_1 < s_2 < \dots < s_k$. Then we have

- $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}$.
- $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}$.
- $\vdots$
- $\min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} \implies X_{k+1} = X \setminus S$.

Thus, $f(k+1) = \min X_{k+1} = x$. Therefore, f is surjective. $\square$

Theorem 2.4.7. *Every infinite subset of a countable set is countable.*

Proof. Let X be a countable set and let $Y \subseteq X$ be an infinite subset. Since X is countable, there exists a bijection $f : \mathbb{N} \rightarrow X$. By [Theorem 2.2.5](#), $f|_{f^{-1}(Y)}^Y : f^{-1}(Y) \rightarrow Y$ is bijective. Then, by [Theorem 2.4.2](#), $f^{-1}(Y) \subseteq \mathbb{N}$ is infinite.

By [Lemma 2.4.6](#), $f^{-1}(Y)$ is countable. Then there exists a bijection $g : \mathbb{N} \rightarrow f^{-1}(Y)$. The composition $f|_{f^{-1}(Y)}^Y \circ g : \mathbb{N} \rightarrow Y$ is bijective, which implies that Y is countable. $\square$

Theorem 2.4.8. *Let X be infinite. If there exists a surjection $f : \mathbb{N} \rightarrow X$, then X is countable.*

Proof. For each $k \in X$, let $S_k := f^{-1}(\{k\})$. Since f is surjective, $S_k \neq \emptyset$ and by Theorem 1.1.1, S_k has a least element for all $k \in X$. Let $g : X \rightarrow \mathbb{N}$ be defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Suppose $g(k_1) = g(k_2)$ for some $k_1, k_2 \in X \implies \min S_{k_1} = \min S_{k_2}$. Since $S_i \cap S_j = \emptyset$ for all $i, j \in X$ with $i \neq j$, we have $S_{k_1} = S_{k_2} \implies k_1 = k_2$. Thus, g is injective.

Since g is injective, by Theorem 2.2.5, $g|^{g(X)} : X \rightarrow g(X)$ is bijective. Since X is infinite, $g(X) \subseteq \mathbb{N}$ is infinite by Theorem 2.4.2. Then, by Lemma 2.4.6, $g(X)$ is countable, which implies that X is countable. $\square$

Theorem 2.4.9. *Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n=1}^{\infty} X_n$ is countable, i.e., any countable union of countable sets is countable.*

Proof. Since each X_n is countable, there exists a bijection $f_n : \mathbb{N} \rightarrow X_n$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we define the followings:

- $t_n = \min T_n$, where $T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\}$.
- $s_n = n - \frac{t_n(t_n-1)}{2} = n - \frac{t_n(t_n+1)}{2} + t_n \in \mathbb{N}$.

Since $2n \in T_n$, $T_n \neq \emptyset$. Then, by Theorem 1.1.1, T_n has a least element. Thus, t_n is well-defined. Since $\frac{t_n(t_n-1)}{2} < n \leq \frac{t_n(t_n+1)}{2}$, we have

$$t_n - s_n + 1 = \frac{t_n(t_n+1)}{2} - n + 1 \geq 1$$

so $t_n - s_n + 1 \in \mathbb{N}$. Let $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ be defined by

$$g(k) = f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$. Then there exists an $m \in \mathbb{N}$ such that $x \in X_m$. Since f_m is surjective, there exists a $l \in \mathbb{N}$ such that $f_m(l) = x$. Let

$$k = \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}.$$

We will show that $l+m-2 \notin T_k$ and $l+m-1 \in T_k$, which implies that

- $t_k = \min T_k = l + m - 1$.
- $s_k = k - \frac{t_k(t_k - 1)}{2} = m$.
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l + m - 1 - m + 1) = f_m(l) = x$.

Consider the following inequalities when $z = l + m - 2$ and $z = l + m - 1$:

- Let $z = l + m - 2$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$

- Let $z = l + m - 1$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1 - l \leq 0. \end{aligned}$$

Thus g is surjective, which implies that $\bigcup_{n=1}^{\infty} X_n$ is countable by [Theorem 2.4.8](#). $\square$

Corollary 2.4.10. *Let $X_1, X_2, \dots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\bigcup_{i=1}^n X_i$ is countable, i.e., any finite union of countable sets is countable.*

Proof. Let $X_{n+1}, X_{n+2}, \dots$ be countable sets. Then, by [Theorem 2.4.9](#), $\bigcup_{n=1}^{\infty} X_n$

is countable. Suppose that $\bigcup_{i=1}^n X_i$ is finite. Then $X_1 \subseteq \bigcup_{i=1}^n X_i$ is finite by [Theorem 2.3.2](#), which contradicts [Theorem 2.4.4](#). Thus, $\bigcup_{i=1}^n X_i$ is infinite. Since

$\bigcup_{i=1}^n X_i \subseteq \bigcup_{n=1}^{\infty} X_n$, by [Theorem 2.4.7](#), $\bigcup_{i=1}^n X_i$ is countable. $\square$

Theorem 2.4.11. *Suppose that X is at most countable, and, for each $x \in X$, Y_x is at most countable. Then the union*

$$Y = \bigcup_{x \in X} Y_x$$

is at most countable, i.e., any at most countable union of at most countable sets is at most countable.

Proof. We have 4 cases to consider:

- (A) Finite union of finite sets is finite by [Theorem 2.3.9](#).
- (B) Finite union of countable sets is countable by [Corollary 2.4.10](#).
- (C) Countable union of finite sets is countable.
- (D) Countable union of countable sets is countable by [Theorem 2.4.9](#).

Let us prove (C). Let X be a countable set and Y_x be finite for each $x \in X$. Then, there exists a bijection $f : \mathbb{N} \rightarrow X$, and for each $x \in X$, there exists a bijection $g_x : J_{n_x} \rightarrow Y_x$ for some $n_x \in \mathbb{N}$. Let $h : \mathbb{N} \rightarrow Y$ be defined by

- $h(j) = g_{f(1)}(j)$ for $j = 1, 2, \dots, n_{f(1)}$.
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) = g_{f(k)}(m)$ for each $k \in \mathbb{N}$ and $m = 1, 2, \dots, n_{f(k)}$.

Then, h is a surjection from $\mathbb{N}$ to Y . By [Theorem 2.4.8](#), Y is countable. $\square$

Lemma 2.4.12. $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. We will explicitly construct a bijection from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be defined by

$$f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

for each $(m, n) \in \mathbb{N} \times \mathbb{N}$. We will show that f is bijective.

- **Injectivity:** Suppose $(m_1, n_1) \neq (m_2, n_2)$. If $m_1 + n_1 = m_2 + n_2$, then $m_1 \neq m_2$, so $f(m_1, n_1) \neq f(m_2, n_2)$. Otherwise, $m_1 + n_1 \neq m_2 + n_2$. Without loss of generality, assume $m_1 + n_1 < m_2 + n_2$. Then, since $m_1 +$

$n_1 + 1 \leq m_2 + n_2$, we have

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2), \end{aligned}$$

Thus, $f(m_1, n_1) \neq f(m_2, n_2)$, so f is injective.

- **Surjectivity:** Let $k \in \mathbb{N}$. Let t_k, s_k be defined as in the proof of [Theorem 2.4.9](#). Then we have $f(s_k, t_k - s_k + 1) = k$. Thus, f is surjective.

□

Theorem 2.4.13. *Let X and Y be countable sets. Then $X \times Y$ is countable.*

Proof. There exist bijections $f : \mathbb{N} \rightarrow X$ and $g : \mathbb{N} \rightarrow Y$. Let $h : \mathbb{N} \times \mathbb{N} \rightarrow X \times Y$ be defined by

$$h(m, n) = (f(m), g(n)) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}.$$

Then, h is a bijection from $\mathbb{N} \times \mathbb{N}$ to $X \times Y$. By [Lemma 2.4.12](#), $\mathbb{N} \times \mathbb{N}$ is countable, so $X \times Y$ is countable. □

Theorem 2.4.14. *Let $X_1, X_2, \dots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\prod_{i=1}^n X_i$ is countable, i.e., any finite product of countable sets is countable.*

Proof. We will prove the theorem by mathematical induction on n . Let $P(n) : \prod_{i=1}^n X_i$ is countable for some $n \in \mathbb{N}$, given that $X_1, X_2, \dots, X_n$ are countable sets.

- **Base case:** Let X_1 be a countable set. Then $\prod_{i=1}^1 X_i = X_1$ is countable. Thus, $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be countable sets. Then we have

$$\prod_{i=1}^{n+1} X_i = \left(\prod_{i=1}^n X_i \right) \times X_{n+1}$$

By the inductive hypothesis, $\prod_{i=1}^n X_i$ is countable. Then, by [Theorem 2.4.13](#),

$\prod_{i=1}^{n+1} X_i$ is countable. Thus, $P(n+1)$ is true when $P(n)$ is true.

By the [principle of mathematical induction](#), $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.4.15. $\mathbb{Q}$ is countable.

Proof. Any $q \in \mathbb{Q}$ can be expressed as $q = \frac{m}{n}$ for some $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Thus, we have

$$f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}, \quad f(m, n) = \frac{m}{n} \text{ for all } (m, n) \in \mathbb{Z} \times \mathbb{N}.$$

is a surjection. By [Theorems 2.4.5](#) and [2.4.13](#), $\mathbb{Z} \times \mathbb{N}$ is countable. Then, there exists a bijection $g : \mathbb{N} \rightarrow \mathbb{Z} \times \mathbb{N}$. Let $h \equiv f \circ g : \mathbb{N} \rightarrow \mathbb{Q}$. By [Theorem 2.2.7](#), h is a surjection.

Since $\mathbb{N} \subseteq \mathbb{Q}$, by [Theorem 2.3.2](#) and [lemma 2.4.3](#), $\mathbb{Q}$ is infinite. Then, by [Theorem 2.4.8](#), $\mathbb{Q}$ is countable. □

Theorem 2.4.16. Let $A := \{0, 1\}$. Then $X = \prod_{n=1}^{\infty} A$ is uncountable.

Proof. For each $n \in \mathbb{N}$, let $e_n \in X$ be the sequence whose n -th term is 1 and all other terms are 0. Let $f : \mathbb{N} \rightarrow X$ be defined by $f(n) = e_n$ for all $n \in \mathbb{N}$. Since $f|_{f(\mathbb{N})}$ is bijective, $f(\mathbb{N}) \subseteq X$ is countable, so is infinite by [Theorem 2.4.4](#).

Then, by [Theorem 2.3.2](#), X is infinite. Suppose that X is countable. Then, there exists a bijection $g : \mathbb{N} \rightarrow X$. Let $X = \{g(1), g(2), g(3), \dots\}$, where

- $g(1) = (a_{11}, a_{12}, a_{13}, \dots)$.
- $g(2) = (a_{21}, a_{22}, a_{23}, \dots)$.
- $\vdots$

Let $x = (b_1, b_2, b_3, \dots) \in X$ be defined by $b_n = 1 - a_{nn}$ for all $n \in \mathbb{N}$. Then, $x \neq g(n)$ for all $n \in \mathbb{N}$, which contradicts the assumption that g is surjective. Thus, X is uncountable. $\square$

Theorem 2.4.17. $\mathbb{R}$ is uncountable.

Proof. Since $\mathbb{N} \subseteq \mathbb{R}$, $\mathbb{R}$ is infinite, as like [Q is infinite](#). Let $f : \mathbb{N} \rightarrow [0, 1)$ be defined by $f(n) = \frac{1}{n+1}$ for all $n \in \mathbb{N}$. $[0, 1) \subseteq \mathbb{R}$ is infinite as like [above](#). We will prove that $[0, 1) \subseteq \mathbb{R}$ is uncountable. Then, by [Theorem 2.4.7](#), $\mathbb{R}$ is uncountable.

Let $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$ be defined by $g(x) :=$ (the [base-2 expansion](#) of x) for all $x \in [0, 1)$. $\square$

2.5 Metric Spaces

2.6 Compact Sets

2.7 Perfect Sets

2.8 Connected Sets

Chapter 3

Numerical Sequences and Series

3.1 Convergent Sequences

3.2 Subsequences

3.3 Cauchy Sequences

3.4 Upper and Lower Limits

3.5 Some Special Sequences

3.6 Series

3.7 Series of Nonnegative Terms

3.8 The Number e

3.9 The Root and Ratio Tests

3.10 Power Series

3.11 Summation by Parts

3.12 Absolute Convergence

3.13 Addition and Multiplication of Series

3.14 Rearrangements

Chapter 4

Continuity

4.1 Limits of Functions

4.2 Continuous Functions

4.3 Continuity and Compactness

4.4 Continuity and Connectedness

4.5 Discontinuities

4.6 Monotonic Functions

4.7 Infinite Limits and Limits at Infinity

Chapter 5

Differentiation

5.1 The Derivative of a Real Function

5.2 Mean Value Theorems

5.3 The Continuity of Derivatives

5.4 L'Hospital's Rule

5.5 Derivatives of Higher Order

5.6 Taylor's Theorem

5.7 Differentiation of Vector-valued Functions

Chapter 6

The Riemann-Stieltjes Integral

6.1 Definition and Existence of the Integral

6.2 Properties of the Integral

6.3 Integration and Differentiation

6.4 Integration of Vector-valued Functions

6.5 Rectifiable Curves

Chapter 7

Sequences and Series of Functions

7.1 Discussion of Main Problem

7.2 Uniform Convergence

7.3 Uniform Convergence and Continuity

7.4 Uniform Convergence and Integration

7.5 Uniform Convergence and Differentiation

7.6 Equicontinuous Families of Functions

7.7 Equicontinuous Families of Functions

7.8 The Stone-Weierstrass Theorem

Chapter 8

Some Special Functions

8.1 Power Series

8.2 The Exponential and Logarithmic Functions

8.3 The Trigonometric Functions

8.4 The Algebraic Completeness of the Complex Field

8.5 Fourier Series

8.6 The Gamma Function

Chapter 9

Functions of Several Variables

9.1 Linear Transformations

9.2 Differentiation

9.3 The Contraction Principle

9.4 The Inverse Function Theorem

9.5 The Implicit Function Theorem

9.6 The Rank Theorem

9.7 Determinants

9.8 Derivatives of Higher Order

9.9 Differentiation of Integrals

Chapter 10

Integration of Differential Forms

10.1 Integration

10.2 Primitive Mappings

10.3 Partitions of Unity

10.4 Change of Variables

10.5 Differential Forms

10.6 Simplexes and Chains

10.7 Stokes' Theorem

10.8 Closed Forms and Exact Forms

10.9 Vector Analysis

Chapter 11

The Lebesgue Theory

11.1 Set Functions

11.2 Construction of the Lebesgue Measure

11.3 Measure Spaces

11.4 Measurable Functions

11.5 Simple Functions

11.6 Integration

11.7 Comparison with the Riemann Integral

11.8 Integration of Complex Functions

11.9 Functions of Class $\mathcal{L}^2$

Bibliography

- [1] Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill International Editions.
- [2] Kenneth Ross, *Elementary Analysis*, Springer Nature.
- [3] Steven Douglass, *Introduction to Mathematical Analysis*, Pearson.
- [4] Terence Tao, *Analysis I*, Springer Nature.